

Institute of Leather Engineering and Technology
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CHAPTER · Wastewater and Solid Waste Management

TERTIARY TREATMENT

Disinfection, Membranes, CETP and Sludge Disposal

A study text written in simple English, arranged topic by topic for revision

*Based on Metcalf & Eddy, Wastewater Engineering: Treatment and Resource Recovery, with Arceivala & Asolekar,
Judd and Crittenden as supporting references*

Contents

Section	Topic	Syllabus item
1	Principles of Tertiary Treatment	—
2	Filtration and Adsorption	prerequisite
3	Disinfection: Principles	—
4	Chlorination	Chlorination
5	Ultraviolet (UV) Treatment	UV treatment
6	Ozonolysis	Ozonolysis
7	Reverse Osmosis and Membrane Processes	Reverse osmosis
8	The Concept of the CETP	Concept of CETP
9	Disposal of Sludge	Disposal of sludge
—	Glossary of Symbols and Abbreviations	

Table 1. The nine topics of this chapter.

1. Principles of Tertiary Treatment

1.1 What secondary treatment leaves behind

Secondary treatment removes the organic matter that bacteria can eat, and it does this well — 85 to 95% of the BOD. But four kinds of pollutant pass straight through it, and these four are the job of tertiary treatment.

- **Fine suspended solids.** The secondary clarifier is good but not perfect. It lets through 20–100 mg/L of small, light biological floc. These particles also carry BOD with them.
- **Colour and hard-to-break organic matter.** Micro-organisms eat only what they can digest. Dyes, syntans and vegetable tannins are not biodegradable. They pass through unchanged and leave the water strongly coloured.
- **Dissolved salts.** Biological treatment cannot remove salt at all. In tannery effluent the TDS is often ten times the limit. This is the worst of the four problems.
- **Pathogens.** Secondary treatment kills 90–99% of the bacteria. That sounds good, but many thousands still remain in every millilitre.

KEY IDEA

Primary treatment removes what settles. Secondary treatment removes what is biodegradable. Tertiary treatment removes what is left — fine solids, colour, salts and pathogens. These are exactly the pollutants that biology cannot touch.

1.2 Definition and position

DEFINITION

Tertiary treatment (also called advanced treatment) is any process used after secondary treatment. It removes the pollutants that biological treatment cannot remove, so that the water meets a strict discharge limit or is clean enough to reuse.

Two things make it different from the stages before it. **It is physical and chemical, not biological** — filtration, adsorption, oxidation, membranes. And **it is selective**: you choose units for the pollutant you actually have. A plant discharging to an irrigation channel may need only disinfection; a plant that wants to reuse its water needs the full membrane train.

1.3 What decides how far a plant must go

Four pressures decide this. They become more demanding as you go down the list.

Driver	What it asks for	Process used
Discharge standard	Colour, COD and solids within the legal limit	Filtration; activated carbon; oxidation
Public health	Kill pathogens before discharge or irrigation	Chlorination, UV or ozone
Water scarcity or cost	Water clean enough to use again in the factory	Ultrafiltration, then reverse osmosis
Zero liquid discharge	No liquid leaves the site at all	RO, evaporation, crystallisation

Table 2. The four drivers of tertiary treatment.

1.4 The discharge limits that apply

In Bangladesh the limits come from the Environmental Conservation Rules 2023 (ECR 2023), for discharge to inland surface water. The ones that matter for tertiary treatment are given below.

Parameter	ECR 2023 limit	Process usually needed
Suspended solids	150 mg/L	Filtration
COD	200 mg/L	Activated carbon or chemical oxidation
BOD₅	50 mg/L	Usually met by secondary; filtration helps
Colour	150 Hazen	Activated carbon; ozone; coagulation
Total dissolved solids	2,100 mg/L	Reverse osmosis, or use less salt at source
pH	6 – 9	Neutralisation
Total chromium	2.0 mg/L	Recovered in primary; polished by ion exchange
Sulphide	2.0 mg/L	Oxidised in primary treatment

Table 3. ECR 2023 limits relevant to tertiary treatment (inland surface water).

Compare a typical tannery with this table. After a good secondary stage the BOD passes, the COD and colour are borderline, and the TDS fails by a wide margin. TDS therefore decides tannery tertiary design, and it is the most expensive to solve.

1.5 The order of the tertiary processes

Tertiary units are almost always placed in a fixed order, because each one protects the next.

Filtration → Adsorption / Oxidation → Membranes → Disinfection

Filtration comes first because suspended solids block carbon beds, foul membranes and hide bacteria from disinfectants. Adsorption and oxidation come next; they remove colour and hard COD, which would otherwise foul the membranes and use up the chlorine or ozone. Membranes then take out the dissolved salts. Disinfection is last, so it acts on water that is already clear and only a small dose is needed.

KEY POINTS — learn these

1. Four pollutants survive secondary treatment: fine suspended solids, colour and refractory organics, dissolved salts, and pathogens.
2. Tertiary treatment is physical and chemical, and it is selective — you choose units for the pollutant you have.
3. Four drivers decide how far to go: discharge standard, public health, water scarcity, zero liquid discharge.
4. The order is filtration → adsorption/oxidation → membranes → disinfection, because each stage protects the next.
5. For a tannery, TDS is the hardest limit to meet and the most expensive to solve.

REMEMBER THESE NUMBERS

Secondary treatment removes **85–95% of BOD** and kills **90–99%** of bacteria.

It leaves **20–100 mg/L** of suspended solids.

ECR 2023: SS **150**, COD **200**, BOD₅ **50 mg/L**; colour **150 Hazen**; TDS **2,100 mg/L**; pH **6–9**; chromium **2.0 mg/L**; sulphide **2.0 mg/L**.

LIKELY EXAM QUESTIONS

1. Name the four kinds of pollutant that survive secondary treatment, and say why biological treatment cannot remove each one.
2. State the four drivers that decide how far tertiary treatment must go, and give the process used for each.
3. Why is filtration placed first in the tertiary sequence? Give three reasons.
4. Which ECR 2023 parameter is hardest for a tannery to meet, and why?

2. Filtration and Adsorption

2.1 Why filtration is needed

A secondary clarifier leaves 20–100 mg/L of suspended solids. These are biological floc particles — light, small and slow to settle — so no extra settling time removes them at sensible cost. They must be strained out. There are three further reasons to do so: they carry BOD, they shelter bacteria from disinfectants, and they foul any membrane or carbon bed placed after them.

2.2 How a granular filter works

Water flows downward through a bed of sand, or of coarse anthracite over sand. The bed is usually 0.6–1.2 m deep. Particles are removed by three mechanisms working together.

- **Straining** — particles bigger than the openings between grains are held at the surface.
- **Sedimentation and interception** — inside the bed, particles settle onto the grains or hit them as the water follows its winding path.
- **Adhesion** — once a particle touches a grain, surface forces hold it there. A thin biological film on the grains helps this.

As the bed fills, the headloss rises and the filtrate gets worse. The bed is then **backwashed**: clean water, often helped by air, is pushed upward to lift the bed and carry the solids away. Backwash water is 2–5% of throughput and returns to the head of the works.

2.3 Types of filter

Type	Description	Where it is used
Rapid gravity sand	Sand bed 0.6–0.9 m deep; 5–15 m ³ /m ² ·h	The standard tertiary filter
Dual or multimedia	Anthracite over sand (over garnet); coarse to fine in the flow direction	Higher solids capacity; longer runs
Pressure filter	Sand in a closed steel vessel under pressure	Small plants; pre-treatment for RO
Disc or cloth filter	Rotating cloth-covered discs; very compact	Retrofit where space is short
Slow sand filter	Fine sand at 0.1–0.2 m ³ /m ² ·h with a biological surface layer	Very high quality; needs a large area

Table 4. Types of tertiary filter.

The dual-media filter often confuses students. In a single sand bed the finest grains settle to the top after backwashing, so all the solids are caught in the top few centimetres and the headloss rises fast. Place coarse, light anthracite above fine, heavy sand and the water meets the coarse media first: solids go deeper, the whole bed is used, and runs are much longer.

2.4 Adsorption on activated carbon

Filtration removes particles. It does nothing about dissolved colour and hard COD, which are the tannery's other typical residues. For those the classical answer is activated carbon.

DEFINITION

Adsorption is the sticking of dissolved molecules onto the **surface** of a solid.

It is not the same as **absorption**, where a substance goes into the body of a material. Dust settling on a wall is adsorption. Water soaking into a sponge is absorption.

Activated carbon is made by charring a carbon-rich material — coconut shell, wood or coal — and then treating it with steam at high temperature. This opens up a huge network of pores inside it. The internal surface area is 500–1,500 m² per gram. One spoonful has about the surface area of a football pitch.

Two properties make carbon good at exactly our pollutants. Its surface is mostly non-polar, so it attracts large non-polar organic molecules — which is what dyes, tannins and syntans are. And its tiny pores give an enormous number of sites for them to stick to.

2.5 Carbon in practice

Property	Powdered (PAC)	Granular (GAC)
Particle size	Less than 0.1 mm	0.6 – 4 mm
How it is used	Dosed into the water, mixed, then filtered off	Packed in a column; water flows through
Contact time	Batch; 30–60 minutes	Continuous; empty-bed contact 10–30 minutes
Regeneration	Usually not regenerated	Heated and reused
Typical use	Occasional or seasonal colour problems	Continuous tertiary polishing

Table 5. Powdered and granular activated carbon compared.

Two limits must be understood. First, the capacity is **finite**. As the carbon fills up, it removes less and less, until it is exhausted. In a column, the moment when the pollutant starts to appear in the outlet is called **breakthrough**. The carbon must then be regenerated or replaced.

Second — and students often miss this — adsorption destroys nothing. The pollution has simply moved from the water onto the carbon. The used carbon is now a concentrated waste, and it must itself be dealt with.

KEY IDEA

Adsorption moves pollution; it does not destroy it. The colour and COD are now on the carbon, and that carbon must be regenerated or dumped. Oxidation processes such as ozonolysis are different — they truly destroy the molecule.

KEY POINTS — learn these

1. Filtration is needed for four reasons: the solids limit, the BOD the solids carry, protection of disinfection, and protection of carbon and membranes.
2. A granular filter removes particles by straining, sedimentation and interception, and adhesion.
3. Dual-media filters run longer because coarse media comes first and the whole depth of the bed is used.
4. Adsorption is sticking onto a surface; absorption is soaking into a body.
5. Carbon capacity is finite; the point when pollutant appears in the outlet is breakthrough.
6. Carbon transfers pollution to a solid waste. Ozone destroys the molecule instead.

REMEMBER THESE NUMBERS

Secondary effluent carries **20–100 mg/L** suspended solids.

Sand bed depth **0.6–1.2 m**; rapid gravity filter rate **5–15 m³/m²·h**; slow sand **0.1–0.2 m³/m²·h**.

Backwash water is **2–5%** of throughput.

Activated carbon internal surface area **500–1,500 m²/g**. GAC empty-bed contact time **10–30 minutes**.

LIKELY EXAM QUESTIONS

1. Give three reasons for filtering a secondary effluent, other than the suspended-solids limit itself.
2. Name and explain the three mechanisms by which a granular filter removes particles.
3. Explain why a dual-media filter gives longer runs than a single-media sand filter.
4. Distinguish adsorption from absorption, with an everyday example of each.
5. What is breakthrough in a carbon column, and why is it said that carbon does not destroy pollution?

3. Disinfection: Principles

3.1 Why disinfection is needed

Secondary treatment kills about 90–99% of bacteria. Raw sewage may hold ten million coliforms per 100 mL, so even after 99% removal about a hundred thousand remain — far too many if the receiving water is used for bathing, irrigation, fishing or supply.

The organisms that matter are bacteria (cholera, typhoid), viruses (hepatitis, polio), protozoan cysts such as *Giardia* and *Cryptosporidium*, and helminth eggs. They differ greatly in how hard they are to kill, and that decides which disinfectant you choose.

3.2 Disinfection is not sterilisation

DEFINITION

Disinfection is the selective destruction of disease-causing organisms — enough to make the water safe. Some organisms still survive.

Sterilisation is the destruction of every organism without exception. It is neither possible nor necessary in wastewater treatment.

3.3 How disinfectants act

A disinfectant kills in one or more of four ways. Knowing which helps explain why the three methods behave differently.

- **Damage to the cell wall** — the contents leak out and the cell dies.
- **Change in cell permeability** — the membrane can no longer control what goes in and out.
- **Damage to nucleic acids** — the organism lives but can no longer reproduce. This is how UV works.
- **Stopping enzyme activity** — the chemistry of the cell is disabled. This is how chlorine and ozone mainly work.

3.4 The governing law: Chick–Watson

Disinfection is not instant. Organisms die gradually, following first-order kinetics. This is Chick's law:

$$dN/dt = -k N \quad \text{or} \quad N_t / N_0 = e^{(-kt)}$$

N = organisms alive at time t ; N_0 = number at the start; k = die-off rate constant

Watson added the effect of concentration. Together they give the rule that governs all disinfection practice:

$$C^n \times t = \text{constant}$$

C = disinfectant concentration; t = contact time; n = dilution coefficient, close to 1

In practice n is taken as 1. The product $C \times t$ — always called the **CT value** — then becomes the single design quantity for disinfection. A given organism needs a given CT for a given level of kill.

KEY IDEA

Disinfection is governed by the CT value: concentration multiplied by contact time. A small dose held for a long time and a large dose held briefly can give the same kill. This is why a disinfection unit is built as a **contact tank**, and why short-circuiting in that tank is so damaging — water that rushes through gets less than its design CT.

3.5 The log-removal convention

The numbers involved cover many orders of magnitude, so performance is written as a logarithm.

$$\text{Log removal} = \log_{10} (N_0 / N)$$

Log removal	Organisms surviving	Percentage killed
1 log	1 in 10	90%
2 log	1 in 100	99%
3 log	1 in 1,000	99.9%
4 log	1 in 10,000	99.99%
6 log	1 in 1,000,000	99.9999%

Table 6. The log-removal convention.

Going from 99% to 99.99% sounds like a gain of only 0.99 percentage points. In fact it is a hundredfold cut in survivors, and it usually needs double the CT value.

3.6 What interferes with disinfection

Six things reduce the effect of any disinfectant. Every one of them matters in a tannery effluent.

Factor	Effect
Suspended solids	Shield organisms and absorb UV light. The most important interference of all — which is why filtration comes first
Organic matter (COD)	Uses up chlorine and ozone chemically, so less is left to act on the organisms
Turbidity and colour	Absorb and scatter UV light, so it cannot penetrate
Ammonia	Reacts with chlorine to form chloramines, which are much weaker disinfectants
pH	Decides which form chlorine takes — hypochlorous acid (strong) or hypochlorite ion (weak)
Temperature	A higher temperature speeds up the kill

Table 7. Factors that interfere with disinfection.

The first two rows explain why disinfection comes **last**. In a turbid, organically loaded effluent the water itself consumes the disinfectant before it reaches the organisms. In a filtered, polished effluent a small dose is enough.

3.7 The three methods side by side

Feature	Chlorination	UV	Ozone
Type of action	Chemical oxidant	Physical (radiation)	Chemical oxidant
Kills by	Stopping enzymes	DNA damage	Breaking the cell wall
Leaves a residual	Yes — its unique merit	No	No
By-products	THMs, chloramines	None	Bromate if bromide present
Effect on colour/COD	Slight	None	Strong — a major bonus
Relative cost	Lowest	Moderate	Highest
Safety	Hazardous gas storage	Safe	Toxic gas; made on site
Needs clear water	Moderately	Critically	Moderately

Table 8. The three disinfection methods compared.

KEY POINTS — learn these

1. Disinfection kills the disease-causing organisms; sterilisation kills all organisms and is not used in wastewater.
2. Disinfectants act by damaging the cell wall, changing permeability, damaging nucleic acids, or stopping enzymes.
3. Chick–Watson gives $C^n t = \text{constant}$. With $n = 1$, the CT value is the design quantity.
4. A contact tank must be plug flow, or part of the water receives less than the design CT.
5. Log removal is $\log_{10}(N_0/N)$. Each extra log is a tenfold cut in survivors.
6. Six interferences: suspended solids, COD, turbidity and colour, ammonia, pH, temperature. Suspended solids matter most.
7. Disinfection goes last because a dirty water consumes the disinfectant before it reaches the organisms.

REMEMBER THESE NUMBERS

Secondary treatment removes **90–99%** of bacteria, leaving about **10^5 coliforms per 100 mL**.

3 log = 99.9%, 4 log = 99.99%, 6 log = 99.9999%.

Going up one log usually needs about **double** the CT value.

Only **chlorine** leaves a residual. Only **ozone** removes colour as well.

LIKELY EXAM QUESTIONS

1. Distinguish disinfection from sterilisation.
2. State the Chick–Watson relationship and explain the meaning of the CT value.
3. If a process gives 3-log removal, what percentage is killed, and how many survive out of one million?
4. List six factors that interfere with disinfection and say which is the most important, and why.
5. Why is disinfection placed last in the tertiary sequence? Which method leaves a residual, and why does that matter?

4. Chlorination

4.1 Why chlorine came first

Chlorine has been used since the early twentieth century and is still the most widely used disinfectant in the world. Three merits explain this: it is cheap, it is effective against bacteria, and — alone among the three methods — it leaves a **residual** that keeps protecting the water after it leaves the plant.

4.2 The chemistry

When chlorine gas dissolves in water it reacts at once to form hypochlorous acid:



Hypochlorous acid then splits up, and how far it splits depends on the pH:



HOCl = hypochlorous acid; OCl⁻ = hypochlorite ion

This is the most important fact in chlorination practice, because the two forms differ greatly in strength. Hypochlorous acid is 40 to 80 times more effective than the hypochlorite ion: HOCl has no charge and passes through the cell wall easily, while OCl⁻ is negative and the cell surface repels it.

Which form you get depends on pH. Below about pH 6 almost all is HOCl; at pH 7.5 the two are equal; above pH 9 almost all is OCl⁻. Together they are the **free available chlorine**.

KEY IDEA

Chlorination works far better at low pH than at high pH, because the strong form is HOCl, not OCl⁻. At pH 6 nearly all the chlorine is in the strong form; at pH 9 almost none of it is. A tannery effluent left alkaline will disinfect very poorly.

4.3 Chloramines and breakpoint chlorination

Tannery effluent contains a lot of ammonia, from deliming and from protein breakdown. Chlorine reacts with ammonia in preference to disinfecting:



These chloramines are called the **combined available chlorine**. They do disinfect, but they are about a hundred times weaker than free chlorine and need much longer contact.

If you keep adding chlorine, the measured residual first rises as chloramines form, then **falls** as more chlorine destroys them, reaching a minimum called the **breakpoint**. Beyond that point any further chlorine appears as free available chlorine. The breakpoint comes at roughly 8–10 parts of chlorine to 1 part of ammonia nitrogen.

Breakpoint chlorination means dosing past this point to get a free residual. For an effluent full of ammonia the demand is therefore very large — one reason why chlorinating tannery effluent is expensive and often unsatisfactory.

4.4 Chlorine demand and dose

Not all the chlorine you add is available to disinfect. Some is used up by organic matter, ammonia, sulphide, iron and other reducing agents in the water. That consumption is the **chlorine demand**.

Dose applied = chlorine demand + residual required

The demand must be measured, never assumed, because it changes with the effluent. For a good secondary effluent it may be 3–8 mg/L. For a tannery effluent with sulphide and ammonia left in it, several times more.

Application	Dose (mg/L)	Contact time
Filtered secondary effluent	5 – 10	30 min at peak flow
Unfiltered secondary effluent	10 – 20	30 – 60 min
Tannery effluent (high demand)	20 – 50 or more	30 – 60 min
Target free residual after contact	0.5 – 1.0	—

Table 9. Indicative chlorine doses and contact times.

4.5 Forms of chlorine used

Form	Description	Notes
Chlorine gas (Cl₂)	Liquefied gas in cylinders	Cheapest per kg; hazardous — needs a chlorine room, leak detection and escape sets
Sodium hypochlorite	Liquid bleach, 10–15% available chlorine	Safer to handle; costs more; loses strength in storage
Calcium hypochlorite	Solid, about 65–70% available chlorine	Convenient for small works; forms deposits
Chlorine dioxide (ClO₂)	Generated on site	Works at high pH; forms no chloramines; costs more

Table 10. Forms in which chlorine is applied.

4.6 The contact tank

Because disinfection depends on the CT value, the contact tank is as much part of the process as the chlorine itself. It is designed to give at least 30 minutes of contact at peak flow.

It must also behave as **plug flow**, so every parcel of water stays the full time. This needs a serpentine baffled channel with a length-to-width ratio of 20:1 or more. Without baffles the tank short-circuits: part of the flow crosses in a fraction of the design time, gets a fraction of the design CT, and the disinfection fails even though the calculation looked correct.

WORKED EXAMPLE — chlorine dose and contact tank

A plant discharges **600 m³/d** of filtered secondary effluent. Peak flow is 1.8 times average. Measured chlorine demand is **12 mg/L**. A free residual of **0.8 mg/L** is wanted after 30 minutes.

Step 1 — Dose. Dose = demand + residual = 12 + 0.8 = 12.8, say **13 mg/L**

Step 2 — Daily consumption. 13 mg/L = 0.013 kg/m³; 0.013 × 600 = **7.8 kg Cl₂ per day**

Step 3 — If 12% hypochlorite is used. 7.8 / 0.12 = 65 kg/d, which at 1.2 kg/L is about **54 litres per day**

Step 4 — Peak flow. 1.8 × 600 = 1,080 m³/d = 45 m³/h = **0.75 m³/min**

Step 5 — Tank volume for 30 min. V = 0.75 × 30 = 22.5 m³, **adopt 25 m³**

Step 6 — Dimensions. Depth 1.5 m → area = 25 / 1.5 = 16.7 m². For 20:1, width² = 16.7/20 = 0.835, so width = **0.91 m** and channel length = **18.3 m**. Arrange as four passes of 4.6 m with baffles.

Comment. The demand (12 mg/L) is fifteen times the residual needed (0.8 mg/L). Almost all the chlorine you buy is used up by the water, not by the organisms. This is why filtration and COD removal *before* chlorination save more money than they cost.

4.7 Disadvantages, and dechlorination

Against its low cost and its residual, chlorination has four serious drawbacks.

- **By-products.** Chlorine reacts with leftover organic matter to form trihalomethanes (THMs) and haloacetic acids, which are suspected carcinogens. A coloured, organically rich tannery effluent is a bad case for this.
- **The residual is toxic.** The same residual that protects the water is poisonous to fish. Where the receiving water is sensitive, the effluent must be **dechlorinated** first, usually with sulphur dioxide or sodium bisulphite. That means another chemical and another cost.
- **Weak against cysts.** Chlorine works well on bacteria but poorly on protozoan cysts such as *Cryptosporidium*.
- **Hazard.** Chlorine gas is acutely toxic. Storing and handling it needs strict safety provision.

KEY POINTS — learn these

1. Chlorine is cheap, effective on bacteria, and the only one of the three that leaves a residual.
2. $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{H}^+ + \text{Cl}^-$, and $\text{HOCl} \rightleftharpoons \text{H}^+ + \text{OCl}^-$. HOCl is the strong form; OCl^- is weak.
3. Low pH favours HOCl, so chlorination is far better at low pH.
4. Free available chlorine = HOCl + OCl^- . Combined available chlorine = the chloramines.
5. Breakpoint chlorination means dosing past the breakpoint to get a free residual.
6. Dose = chlorine demand + residual required. The demand must be measured.
7. The contact tank must be baffled to give plug flow, or the CT is not delivered.
8. Four drawbacks: THMs, a toxic residual, poor kill of cysts, and the hazard of the gas.

REMEMBER THESE NUMBERS

HOCl is **40–80 times** stronger than OCl^- . At **pH 6** almost all HOCl; at **pH 7.5** about equal; at **pH 9** almost all OCl^- .

Chloramines are about **100 times weaker** than free chlorine.

Breakpoint comes at about **8–10 parts Cl_2 to 1 part ammonia-N**.

Typical doses: filtered secondary **5–10 mg/L**; tannery **20–50 mg/L**. Target residual **0.5–1.0 mg/L**.

Contact time **at least 30 minutes at peak flow**; length-to-width ratio **20:1 or more**.

LIKELY EXAM QUESTIONS

1. Write the hydrolysis of chlorine in water and explain why pH matters so much.
2. Distinguish free available chlorine from combined available chlorine.
3. What is breakpoint chlorination, and why is the breakpoint dose so high for tannery effluent?
4. Define chlorine demand and state how the applied dose is decided.
5. Why must a chlorine contact tank be baffled? Give two reasons why an effluent may need dechlorination.

5. Ultraviolet (UV) Treatment

5.1 The problem it solves

Chlorination is cheap and effective, but it adds a chemical, forms by-products, and leaves a residual toxic to aquatic life. Suppose we could disinfect by purely physical means, adding nothing at all. That is what UV does.

5.2 The idea in plain words

Sunlight has always disinfected water, because its ultraviolet part damages micro-organisms. UV treatment concentrates that effect: it makes light of exactly the wavelength most harmful to living cells, and passes the water close to the lamps so every organism gets a lethal exposure. Nothing is added and nothing is left behind. The organisms are not poisoned — they are simply made unable to reproduce.

5.3 How it works

UV light kills best at a wavelength of about **254 nm**. This is very close to the wavelength that DNA and RNA absorb most strongly. When a photon at 254 nm is absorbed by the DNA of a micro-organism, it makes two neighbouring thymine bases bond together. The result is called a **thymine dimer**.

The consequence is decisive. The DNA can no longer be read or copied, so the organism cannot multiply. It may stay alive for a while, but it cannot cause infection. In disinfection terms it is dead.

KEY IDEA

UV does not poison the organism. It damages the DNA so the organism cannot reproduce. This is why UV is called **inactivation** rather than killing — and why photoreactivation, described below, is a real concern.

5.4 Equipment

A UV unit is a set of lamps inside quartz sleeves. They sit either in a closed pressure vessel or in an open channel through which the effluent flows. Quartz is used because ordinary glass absorbs UV.

Lamp type	Characteristics	Where used
Low-pressure	One wavelength at 254 nm; efficient; long life (8,000–12,000 h)	The standard choice for wastewater
Low-pressure high-output	More output per lamp, so fewer lamps needed	Medium and large plants
Medium-pressure	Many wavelengths, high intensity; shorter life; more power	Where space is very restricted

Table 11. UV lamp types.

5.5 The UV dose

The dose is intensity multiplied by exposure time. It is the direct equivalent of the CT value.

$$\text{UV dose} = I \times t$$

I = average intensity (mW/cm²); *t* = exposure time (s); dose in mJ/cm²

Requirement	Dose (mJ/cm ²)	Comment
Typical wastewater disinfection	20 – 40	Standard design range
Water reuse (strict)	40 – 100	Higher assurance needed
4-log virus inactivation	about 100 – 186	Viruses are very resistant
Protozoan cysts	10 – 20	UV is very effective here — unlike chlorine

Table 12. Indicative UV doses.

The last row deserves emphasis. Cysts such as *Cryptosporidium* resist chlorine strongly but are easily inactivated by UV. Where such organisms are the concern, UV is clearly the better method.

5.6 Transmittance: the critical constraint

UV works only if the light reaches the organism. How well it does so is measured by the **UV transmittance (UVT)** — the percentage of 254 nm light that passes through 1 cm of the water.

UVT at 254 nm	Water quality	Consequence
Above 65%	Good filtered effluent	UV works well; normal design
55 – 65%	Moderate	More lamps required

UVT at 254 nm	Water quality	Consequence
Below 50%	Poor — coloured or turbid	UV becomes uneconomic or simply fails

Table 13. Effect of UV transmittance.

Here is the central difficulty for the leather industry. Tannery effluent is **coloured**, and colour absorbs UV strongly. A dark effluent may have a transmittance of only 20–30%, at which UV is not viable at all. Suspended solids make it worse, both by absorbing light and by hiding organisms in their shadow. For a coloured tannery effluent, UV can only be considered **after** the colour has been removed.

5.7 Photoreactivation and dark repair

Some bacteria can repair the damage. Given visible light, an enzyme can split the thymine dimers again and restore the organism. This is **photoreactivation**. A slower repair without light is called **dark repair**.

The designer allows for this by applying a dose larger than the minimum, so that the damage is too great to repair.

5.8 Advantages and limitations

Advantages	Limitations
No chemicals added; nothing left in the water	No residual protection at all
No disinfection by-products	Very sensitive to turbidity, colour and solids
Very effective against protozoan cysts	Photoreactivation can revive some organisms
Very short contact time (seconds)	Lamps and sleeves need regular cleaning
Compact; no hazardous chemical stored	Lamps must be replaced periodically
Simple and safe to operate	Continuous electricity demand

Table 14. Ultraviolet treatment — advantages and limitations.

KEY POINTS — learn these

1. UV inactivates rather than kills: it damages DNA so the organism cannot reproduce.
2. The germicidal peak is 254 nm, close to the absorbance peak of nucleic acids.
3. The damage is a thymine dimer — two neighbouring thymine bases bonded together.
4. UV dose = intensity × time, and it is the direct equivalent of the CT value.
5. UV transmittance decides whether UV is possible. Colour and solids destroy it.
6. UV leaves no residual and forms no by-products, but photoreactivation can restore some organisms.
7. UV beats chlorine on protozoan cysts; chlorine beats UV where a residual is needed.

REMEMBER THESE NUMBERS

Germicidal wavelength **254 nm**. Lamp life **8,000–12,000 hours**.

Typical wastewater dose **20–40 mJ/cm²**; reuse **40–100**; 4-log virus **100–186**; cysts **10–20**.

UVT above **65%** is good; below **50%** UV fails. Tannery effluent is often only **20–30%**.

LIKELY EXAM QUESTIONS

1. At what wavelength is UV most germicidal, and why that wavelength?
2. Explain what a thymine dimer is and why it inactivates an organism.
3. Define UV dose and state the typical range for wastewater disinfection.
4. What is UV transmittance, and why does it make UV difficult for tannery effluent?
5. Explain photoreactivation and how the designer allows for it. Give two cases where UV is preferred to chlorine, and one where it is not.

6. Ozonolysis

6.1 The problem it solves

Chlorine leaves by-products. UV cannot get through a coloured water and does nothing to the colour itself. Ozone answers both objections: it is a far stronger oxidant, it destroys colour and hard organic matter as well as micro-organisms, and it breaks back down into oxygen, leaving no residue.

The word **ozonolysis** means the splitting of organic molecules by ozone. That is exactly why ozone removes colour: it breaks the double bonds of the chromophores that make dye molecules coloured.

6.2 The chemistry

Ozone is an unstable three-atom form of oxygen, O_3 . It cannot be stored, so it must be made on site. It attacks organic matter and micro-organisms in two ways.

- **Direct attack.** The ozone molecule itself reacts selectively with unsaturated bonds — the carbon–carbon double bonds of dyes and aromatics. This is the cleavage that gives ozonolysis its name.
- **Indirect attack.** At higher pH, ozone breaks down in water to form **hydroxyl radicals** ($\cdot OH$). These are even stronger oxidants and attack almost any organic molecule, without selectivity.



The hydroxyl radical has the second-highest oxidation potential of any species usable in water treatment: **2.80 V**, against 1.36 V for chlorine. This is the basis of the **advanced oxidation processes**, in which ozone is deliberately combined with hydrogen peroxide or UV to make as many radicals as possible.

6.3 How ozone is made and applied

Ozone is produced by passing dry air or oxygen through a high-voltage electrical discharge, called the corona discharge. This splits oxygen molecules, which then recombine as ozone:



Yields are 1–3% from air and 6–14% from pure oxygen. The gas is dissolved into the wastewater in a contact column, usually through fine-bubble diffusers at the bottom of a deep tank, with 10–20 minutes of contact. Because ozone is toxic, the off-gas must pass through a destructor before release.

6.4 Doses and performance

Purpose	Dose (mg/L)	Notes
Disinfection of clean effluent	5 – 15	Contact 10–15 min
Colour removal	20 – 50	Often the governing duty for tannery effluent
Refractory COD reduction	30 – 100+	Cost rises steeply with COD
Advanced oxidation ($O_3 + H_2O_2$)	variable	For persistent micro-pollutants

Table 15. Indicative ozone doses.

The second row is why ozone interests the leather industry. A tannery rarely chooses ozone as a disinfectant, because the effluent seldom needs that level of pathogen control. But it is a serious candidate for **colour removal**, where the alternative is activated carbon.

6.5 Ozone against activated carbon for colour

Aspect	Activated carbon	Ozone
Mechanism	Adsorption — moves the pollutant	Oxidation — destroys the molecule
Residue	Spent carbon must be regenerated or dumped	None; ozone turns back into oxygen
Capacity	Finite; breakthrough happens	Continuous; never exhausted
Capital cost	Lower	High (generator, contactor, destructor)
Running cost	Carbon replacement or regeneration	Electricity — substantial
Side benefit	Also removes COD	Makes the remaining organics more biodegradable
Risk	Carbon disposal is a waste problem	Bromate forms if bromide is present

Table 16. Colour removal: carbon adsorption compared with ozonation.

One more advantage deserves note. By cutting large refractory molecules into smaller pieces, ozonation often makes the remaining organic matter **biodegradable**. An ozone stage between two biological stages therefore turns hard COD into food for the second stage — a combination that treats what neither process could handle alone.

6.6 Advantages and limitations

Advantages	Limitations
Very powerful; works on all organism types	Highest capital and energy cost of the three
Destroys colour and hard COD as well as pathogens	No residual protection
Leaves no chemical residue — turns back into oxygen	Must be made on site; cannot be stored
No chlorinated by-products	Forms bromate where bromide is present
Makes residual organics more biodegradable	Toxic gas; an off-gas destructor is essential
Adds dissolved oxygen to the effluent	Needs skilled operation and maintenance

Table 17. Ozonolysis — advantages and limitations.

WORKED EXAMPLE — ozone for colour removal

A tannery discharges **500 m³/d** of biologically treated effluent. Bench trials show a dose of **35 mg/L** brings the colour below 150 Hazen.

Step 1 — Ozone per day. 35 mg/L = 0.035 kg/m³; 0.035 × 500 = **17.5 kg O₃/d**

Step 2 — Allow for transfer. At 90% transfer, generator output = 17.5 / 0.90 = 19.4, say **20 kg/d** = 0.83 kg/h

Step 3 — Power. From oxygen feed at 10 kWh/kg: 0.83 × 10 = **8.3 kW**, about **200 kWh/day**

Comment. Compare this with the chlorination example, where 7.8 kg/d of chlorine was enough. Ozone is doing a harder job — destroying colour, not just killing organisms — and the energy cost shows it. The design question is always whether that cost is less than the cost of the carbon it replaces.

6.7 Choosing between the three methods

Situation	Preferred method
Discharge far away; a residual is wanted	Chlorination
Sensitive receiving water; no chemicals allowed	UV — if the effluent is clear
Effluent is coloured and colour must go too	Ozone
Protozoan cysts are the concern	UV
Ammonia is high	UV or ozone — chlorine would form chloramines
Budget is the overriding constraint	Chlorination
Water is to be reused after RO	UV, usually as a final barrier

Table 18. Selecting a disinfection method.

KEY POINTS — learn these

1. Ozone is O₃, unstable, and must be generated on site because it cannot be stored.
2. It attacks directly at double bonds, and indirectly through hydroxyl radicals at higher pH.
3. It removes colour because it breaks the double bonds of the chromophores in dye molecules.
4. Carbon transfers pollution; ozone destroys it. Carbon has finite capacity; ozone does not.
5. Ozone makes leftover organics biodegradable, so it works well between two biological stages.
6. Ozone is the most expensive of the three and leaves no residual, but it is the only one that removes colour.

REMEMBER THESE NUMBERS

Hydroxyl radical **2.80 V**; chlorine **1.36 V**.

Ozone yield **1–3% from air**, **6–14% from oxygen**. Contact time **10–20 min**.

Doses: disinfection **5–15 mg/L**; colour removal **20–50 mg/L**; COD reduction **30–100+ mg/L**.

Energy **15–20 kWh/kg O₃ from air**, **8–12 kWh/kg from oxygen**.

LIKELY EXAM QUESTIONS

1. Why must ozone be generated on site?
2. Distinguish the direct and the indirect (radical) mechanisms of ozone attack.
3. Explain why ozonolysis removes colour, in terms of what it does to a dye molecule.
4. Compare ozone with activated carbon for colour removal under four headings.
5. Why may an ozone stage placed between two biological stages be particularly effective?

7. Reverse Osmosis and Membrane Processes

7.1 The problem: dissolved salts

We now come to the problem that nothing so far can touch. Screening, settling, biological treatment, filtration, adsorption and oxidation all leave the **dissolved salts** exactly as they found them.

For the leather industry this is the central difficulty. Salt is used freely — curing, pickling, chrome tanning — and all of it ends up in the effluent. Tannery effluent commonly carries 10,000–40,000 mg/L of TDS against a limit of 2,100 mg/L, so even a perfectly run biological plant fails on TDS alone. Only two answers exist: do not use the salt in the first place (low-salt or salt-free curing), or separate it physically with a membrane.

7.2 The idea in plain words

Put fresh water and salt water on either side of a membrane that lets water through but not salt. Water moves by itself from the fresh side to the salty side, diluting it. This is **osmosis**. The pressure that would just stop that flow is the **osmotic pressure** of the salt solution.

Now apply to the salty side a pressure **greater** than the osmotic pressure. The flow reverses. Water is squeezed out of the salt solution and through the membrane, leaving the salt behind. This is **reverse osmosis**.

It is not filtration in the ordinary sense: the membrane has no pores that salt is too big to enter. Water dissolves into the dense polymer film and diffuses through it.

KEY IDEA

Reverse osmosis is the only practical way to remove dissolved salt from wastewater. It works by applying a pressure greater than the osmotic pressure, forcing water through a dense membrane and leaving the salts behind.

7.3 The family of membrane processes

Process	Pore size / cut-off	Removes	Pressure
Microfiltration (MF)	0.1 – 10 μm	Suspended solids, bacteria	0.1 – 2 bar
Ultrafiltration (UF)	0.01 – 0.1 μm	Colloids, macromolecules, viruses	1 – 5 bar
Nanofiltration (NF)	about 0.001 μm	Divalent ions, colour, some salts	5 – 20 bar
Reverse osmosis (RO)	non-porous	Essentially all dissolved salts	15 – 80 bar

Table 19. The membrane family. Only NF and RO remove dissolved salts.

Two points follow. MF and UF do not desalinate at all; their job is to protect the RO by removing what would foul it. And nanofiltration is a useful middle option: it rejects divalent ions such as sulphate and the larger coloured molecules, while passing much of the single-charged salt, at a fraction of RO's pressure.

7.4 Osmotic pressure and operating pressure

For practical work a simple approximation is enough:

$$\pi \approx 0.077 \times \text{TDS} / 1000 \quad (\text{bar, with TDS in mg/L})$$

about 0.077 bar for every 1,000 mg/L of TDS

The applied pressure must be well above this, because the driving force for flow is the excess of applied pressure over osmotic pressure:

$$J = A (\Delta P - \Delta \pi)$$

J = water flux; A = membrane permeability; ΔP = applied pressure; $\Delta \pi$ = osmotic pressure difference

7.5 Recovery, rejection and the concentrate

Three quantities describe how an RO plant works. You should be able to define all three.

- **Recovery** — the fraction of the feed that comes out as clean permeate. Typically 50–75% for a wastewater RO. The rest leaves as concentrate.
- **Rejection** — the percentage of a dissolved substance held back by the membrane. Typically 95–99% for common salts.
- **Concentrate** (also called reject or brine) — the stream carrying the rejected salts, at a much higher concentration.

$$\text{Recovery (\%)} = \text{permeate flow} \div \text{feed flow} \times 100$$

$$\text{Rejection (\%)} = (1 - \text{permeate concentration} \div \text{feed concentration}) \times 100$$

KEY IDEA

Reverse osmosis does not destroy salt — it **concentrates** it. A plant recovering 70% of its water makes a concentrate about three times as salty as the feed, and that stream must then be dealt with. The concentrate is the central difficulty of every RO scheme.

WORKED EXAMPLE — reverse osmosis mass balance

Feed **400 m³/d** at **12,000 mg/L** TDS. Recovery **70%**, rejection **97%**.

Step 1 — Flows. $Q_p = 0.70 \times 400 = 280 \text{ m}^3/\text{d}$; $Q_c = 400 - 280 = 120 \text{ m}^3/\text{d}$

Step 2 — Permeate quality. $C_p = 12,000 \times (1 - 0.97) = 360 \text{ mg/L}$ — well below the 2,100 mg/L limit, and fit for reuse.

Step 3 — Salt balance. Salt in feed = $400 \times 12,000 = 4,800,000 \text{ g/d}$. Salt in permeate = $280 \times 360 = 100,800 \text{ g/d}$. Salt in concentrate = $4,699,200 \text{ g/d}$. $C_c = 4,699,200 / 120 = 39,160 \text{ mg/L}$ — three times the feed and nineteen times the discharge limit.

Step 4 — Osmotic pressures. Feed: $0.077 \times 12 = 0.92 \text{ bar}$. Concentrate: $0.077 \times 39.16 = 3.0 \text{ bar}$.

Step 5 — Operating pressure. It must exceed the osmotic pressure at the concentrate end, and also provide the driving force. In practice **15–25 bar**.

Comment. The plant has recovered 70% of its water, which was the object. But it has also made 120 m³/d of a very salty stream that cannot be discharged and cannot be treated biologically.

7.6 Fouling: the operating problem

Every RO plant lives or dies by the control of fouling. Four kinds occur.

- **Particulate fouling** — suspended matter settling on the membrane. Prevented by MF or UF pre-treatment.
- **Scaling** — sparingly soluble salts, mainly calcium sulphate and carbonate, precipitating as they concentrate. Controlled by antiscalant dosing and by limiting recovery.
- **Organic fouling** — a film of organic matter sticking to the surface. Controlled by removing COD and colour first, which is why carbon or ozone comes before RO.
- **Biofouling** — bacteria growing on the membrane as a biofilm. Controlled by disinfecting the feed. Note that free chlorine damages polyamide membranes and must be removed before the membrane.

The lesson is always the same: RO needs a thoroughly pre-treated feed. A typical reuse train reads:

secondary effluent → filtration → carbon or ozone → ultrafiltration → antiscalant → RO

7.7 The concentrate problem and zero liquid discharge

The worked example produced 120 m³/d of concentrate at nearly 40,000 mg/L. This stream cannot be discharged, cannot be treated biologically, and cannot be ignored. Four options exist.

- Discharge to sea or a deep well, where geography and law allow. Rarely available in Bangladesh.
- Evaporation ponds, where land and climate allow. Simple, but needs a large area and secure lining.
- Mechanical evaporation and crystallisation, producing a solid salt. This is the zero-liquid-discharge route.
- Reduce the salt at source, so that less arrives to be concentrated.

Zero liquid discharge (ZLD) means that no liquid effluent leaves the site at all. The train is: RO to recover most of the water, then an evaporator to boil off more, then a crystalliser to dry the residue to a solid salt.

ZLD is technically complete and has been made compulsory for tannery clusters in parts of India. But it is very expensive: evaporation uses a great deal of energy, the plant is complex, and the mixed salt recovered is usually too impure to sell. It is adopted where regulation compels it, or where water is so scarce that recovery pays.

KEY IDEA

Every kilogram of salt that enters the tannery must eventually leave it — in the effluent, in the concentrate, or as solid salt from a crystalliser. The cheapest place to deal with it is always the curing stage, before it is ever dissolved.

7.8 Advantages and limitations of RO

Advantages	Limitations
The only practical way to remove dissolved salts	High capital and energy cost (15–80 bar pumping)
Produces very high quality water, fit for reuse	Produces a concentrate that must be disposed of
Also removes colour, COD, metals and pathogens	Extensive pre-treatment is essential
Modular — capacity easily extended	Membranes foul and need chemical cleaning
Allows recovery of both water and chemicals	Membranes replaced every 3–7 years
Compact for the duty performed	Needs skilled operation and monitoring

Table 20. Reverse osmosis — advantages and limitations.

KEY POINTS — learn these

1. Only membranes can remove dissolved salt. Nothing else in the treatment train touches it.
2. Osmosis: water moves to the salty side. Reverse osmosis: apply more than the osmotic pressure and the flow reverses.
3. MF and UF do not desalinate; they protect the RO. NF removes divalent ions and colour. RO removes everything.
4. Recovery = permeate ÷ feed. Rejection = the percentage of salt held back.
5. RO concentrates salt; it does not destroy it. The concentrate is the central problem.
6. Four kinds of fouling: particulate, scaling, organic, biological.
7. ZLD = RO → evaporator → crystalliser. Complete, but very expensive.

REMEMBER THESE NUMBERS

Tannery TDS **10,000–40,000 mg/L**; limit **2,100 mg/L**.

Pressures: MF **0.1–2**, UF **1–5**, NF **5–20**, RO **15–80 bar**.

Osmotic pressure $\approx$ **0.077 bar per 1,000 mg/L TDS**.

Recovery **50–75%**; rejection **95–99%**; membrane life **3–7 years**.

LIKELY EXAM QUESTIONS

1. Explain osmosis and reverse osmosis, and define osmotic pressure.
2. Which membrane processes remove dissolved salts and which do not? What is the role of those that do not?
3. Define recovery and rejection, and write the expression for each.
4. Name the four kinds of RO fouling and state how each is controlled.
5. What is zero liquid discharge, and why is it so expensive? Where is the cheapest place to deal with salt?

8. The Concept of the Common Effluent Treatment Plant

8.1 The problem

Everything so far assumes a factory big enough to build and staff its own treatment works. For much of the leather industry that is false. A tannery processing a few hundred hides a day cannot justify a full plant, and cannot afford the land, the capital, or a trained chemist and operator.

Yet the effluent of fifty such tanneries, discharged untreated, is just as damaging as that of one large one. The problem is not technical but structural: how can small firms, each unable to treat its effluent alone, do so together?

8.2 The idea

DEFINITION

A **Common Effluent Treatment Plant (CETP)** is a single treatment works serving a cluster of industrial units. Each unit sends its effluent through a common sewer, and the capital and running costs are shared in proportion to how much each one discharges.

The logic is that of any shared utility. One plant of 5,000 m³/d costs far less to build and run than fifty of 100 m³/d, and it can afford what none of them could: proper equalisation, a laboratory, trained staff, monitoring and sludge handling. It also gives the regulator one point of compliance instead of fifty.

8.3 Advantages of the CETP

- **Economy of scale.** Cost per cubic metre falls sharply with size, and so does the cost of chemicals and sludge disposal.
- **Professional operation.** One plant can employ qualified staff and run a laboratory. No small tannery can.
- **Better equalisation.** Combining many discharges evens out the peaks. One tannery's lime dump is diluted by forty others' washings.
- **Enforceable monitoring.** One outfall can be metered and sampled continuously.
- **Land efficiency.** One site instead of fifty, with room for the tertiary and sludge facilities individual units would leave out.
- **Advanced treatment becomes possible.** RO, ZLD and chrome recovery become affordable at cluster scale.

8.4 Why CETPs often fail

It must be said plainly that many CETPs in South Asia have performed poorly. Students should know why, because the reasons are instructive and largely avoidable.

Cause	Explanation
Hydraulic overload	The cluster grows beyond the design capacity; more tanneries connect than were planned for
No pre-treatment at source	Chrome and sulphide arrive mixed in the common sewer, where neither can be recovered or oxidised properly
Highly variable influent	Fifty tanneries running different processes produce a mixture no single design can suit
Illegal or unmetered discharge	Units bypass the sewer or dump slugs at night; charging by production instead of measured flow removes any reason to save water
Weak cost recovery	Members do not pay; maintenance is deferred; equipment fails
Sludge neglected	The chrome-bearing sludge is hazardous, and safe disposal is expensive and often skipped
Diffused responsibility	When many parties share a plant, no single party feels responsible for it

Table 21. Common causes of CETP failure.

KEY IDEA

A CETP can only be as good as the pre-treatment done at the tanneries feeding it. Chrome recovery, sulphide oxidation and screening **must** be done at source. Once these streams mix in the common sewer, neither the chromium nor the sulphide can be dealt with properly, and the sludge becomes hazardous waste.

8.5 Compulsory pre-treatment at source

It follows that a well-designed CETP scheme always puts obligations on its members. The minimum requirements are:

- **Screening** of coarse solids, so hair, trimmings and fleshings do not enter the sewer.

- **Chrome segregation and recovery** at each tannery, so the common sludge is not contaminated.
- **Sulphide oxidation** in the beamhouse stream before discharge, so hydrogen sulphide is not generated in the sewer — a real danger to sewer workers, not merely a nuisance.
- **Equalisation and pH correction** within limits, so that slugs are not passed on.
- **Metering** of the discharge, so charges follow the actual load and saving water is rewarded.

8.6 The design of a CETP

In its unit processes a CETP is simply a large tannery treatment plant. Everything in this chapter and the last applies. The typical train is:

Screening → Equalisation → Sulphide oxidation → Coagulation → Primary clarifier → Biological stage → Secondary clarifier → Tertiary polishing → (RO / ZLD)

Three things make it different from a single-factory plant. Equalisation must be generous, because the influent is the sum of many uncoordinated discharges. Monitoring must be continuous at both inlet and outlet, for compliance and for charging. And sludge handling must be built on a scale that individual members would never provide.

8.7 The Bangladesh experience

The move of the tannery industry from Hazaribagh to the Savar Tannery Estate, with a CETP at its centre, is the obvious case to study. It shows both the promise and the difficulties of the approach.

The promise was real: a modern plant serving the whole cluster, ending the untreated discharge that had made the Buriganga one of the most polluted rivers in the country. The difficulties met were exactly those in the table above — capacity outpaced by the units connected, pre-treatment not enforced so chromium and sulphide arrive mixed, and above all the disposal of chrome-bearing sludge.

The lesson is not that the CETP idea is unsound. It is that a CETP is a **system**, of which the plant is only one part. The sewer, the pre-treatment at each factory, the metering, the charging structure and the enforcement are parts of it too, and the system performs no better than its weakest part.

KEY POINTS — learn these

1. A CETP is one plant serving a cluster, with shared costs and a common sewer.
2. It exists to solve a structural problem, not a technical one: small firms cannot treat effluent alone.
3. Advantages: economy of scale, professional operation, better equalisation, enforceable monitoring, land efficiency, affordable advanced treatment.
4. Common failures: overload, no pre-treatment at source, variable influent, illegal discharge, weak cost recovery, neglected sludge, diffused responsibility.
5. Chrome recovery and sulphide oxidation must happen at each tannery, never at the CETP.
6. Charging by measured flow rewards water saving; charging by production does not.
7. A CETP is a system, not a plant.

REMEMBER THESE NUMBERS

One plant of **5,000 m³/d** is far cheaper than fifty of **100 m³/d**.

Five compulsory duties at source: **screening, chrome recovery, sulphide oxidation, equalisation and pH control, metering**.

The standing case study is **Hazaribagh → Savar**.

LIKELY EXAM QUESTIONS

1. Define a CETP and explain the structural problem it exists to solve.
2. Give four advantages of a CETP over individual treatment plants.
3. List five common causes of CETP failure.
4. Why must chrome recovery and sulphide oxidation be done at each tannery rather than at the CETP?
5. "A CETP is a system, not a plant." Explain, using the Savar experience.

9. Disposal of Sludge

9.1 The problem

Every process in this chapter has been described as removing pollutants from water. None of them destroys those pollutants, except partly in oxidation. What they do is **concentrate** them into a much smaller volume of solids: sludge.

This is easily forgotten. A treatment plant does not make pollution disappear; it moves it from a large volume of water into a small volume of sludge, and that sludge becomes the plant's biggest remaining problem. Sludge handling routinely takes a third to a half of total operating cost.

KEY IDEA

Wastewater treatment does not destroy pollution. It concentrates it into sludge. Everything removed from the water is still on the site and must still be dealt with. In a tannery, that sludge contains the chromium.

9.2 Sources and quantities of sludge

Source	Nature	Dry solids	Character
Screenings	Hair, trimmings, fibre	—	Handled as solid waste
Primary sludge	Settled solids and chemical floc	2 – 6%	Bulky; easy to dewater
Chrome sludge	Cr(OH) ₃ from chrome recovery	5 – 15%	Recoverable as tanning liquor
Biological sludge	Surplus micro-organisms	0.5 – 2%	Very watery; hard to dewater
Chemical sludge	From coagulation or precipitation	1 – 3%	Voluminous

Table 22. Sources of sludge in a tannery treatment plant.

Look at the dry-solids column. A biological sludge at 1% solids is 99% water: one tonne holds ten kilograms of sludge and 990 kilograms of water. Almost all of sludge handling is therefore about removing water, because it is water, not solids, that must be carted away.

9.3 The sludge treatment sequence

Sludge processing follows a standard order. Each stage removes more water than the last.

Thickening → Stabilisation → Conditioning → Dewatering → Disposal

(a) Thickening

Raises the solids from about 1% to 4–6%, cutting the volume by three-quarters or more before anything else is done. Gravity thickeners suit primary sludge; dissolved-air flotation or mechanical thickeners suit the lighter biological sludge.

Going from 1% to 5% solids cuts the volume to one fifth. Since every later cost depends on volume, thickening is by far the most cost-effective step in the sequence.

(b) Stabilisation

Raw sludge rots and smells. Stabilisation reduces the biodegradable content so the sludge can be handled and stored without nuisance. The methods are anaerobic digestion (which also gives biogas), aerobic digestion, and adding lime to raise the pH above 12 and stop biological activity.

For tannery sludge, anaerobic digestion is difficult because chromium and sulphide poison the methane-forming bacteria. Where the plant runs at extended aeration the sludge is already largely stabilised, so a separate digestion stage is often left out.

(c) Conditioning

Chemical or thermal treatment that makes the sludge give up its water more easily. Usually a polyelectrolyte is dosed. It flocculates the fine particles into larger lumps from which water drains freely. Lime and ferric chloride are also used.

(d) Dewatering

The decisive step. It turns a liquid sludge into a handleable solid cake.

Method	Cake solids	Area / power	Comment
Drying beds	25 – 40%	Large area, no power	Cheapest; suits warm dry climates; rain interrupts
Belt filter press	18 – 25%	Moderate	Continuous; moderate cost; widely used
Plate-and-frame press	30 – 45%	Moderate	Driest cake; batch; best for chrome sludge
Centrifuge	20 – 30%	High power	Compact and enclosed; higher running cost
Screw press	18 – 25%	Low power	Simple, slow, low maintenance

Table 23. Sludge dewatering methods.

For tannery sludge the plate-and-frame press is often preferred. The drier cake reduces both the volume and the cost of hazardous-waste disposal, and batch operation suits the intermittent production of chrome sludge.

WORKED EXAMPLE — the arithmetic of dewatering

A plant produces **250 kg/d** of dry sludge solids from its biological stage at **1.0%** concentration.

As produced (1.0%). $1.0\% = 10 \text{ kg dry solids per m}^3$. Volume = $250 / 10 = \mathbf{25 \text{ m}^3/d}$ (99.0% water)

After thickening to 5%. $5\% = 50 \text{ kg/m}^3$. Volume = $250 / 50 = \mathbf{5 \text{ m}^3/d}$ (95.0% water) — one fifth of the original

After dewatering to 30% cake. $30\% = 300 \text{ kg/m}^3$. Volume = $250 / 300 = \mathbf{0.83 \text{ m}^3/d}$ (70% water), about **833 kg/d of cake**

Comment. The dry solids never changed — 250 kg/d throughout. All that has been removed is water. Yet the volume to be carted away has fallen thirtyfold. If disposal costs Tk 2,000 per m^3 , the difference between hauling 25 m^3 and 0.83 m^3 a day is about Tk 48,000 every day. This single calculation explains why dewatering equipment is worth its capital cost.

9.4 Disposal and utilisation routes

Route	Requirements	Suitability for tannery sludge
Sanitary landfill	Lined site with leachate collection; dewatered cake	The usual route; must be a secure landfill if it carries chrome
Agricultural use	Low metals; stabilised; free of pathogens	Generally not permitted for chrome sludge
Incineration	High capital; air-pollution control	Cuts volume greatly, but risks oxidising Cr(III) to Cr(VI)
Cement kiln co-processing	Consistent quality; agreement with the kiln	Promising: high temperature, chromium locked into the clinker
Building materials	Bricks, blocks, aggregate	Under research; leaching must be proved acceptable
Chromium recovery	Segregated chrome sludge	The preferred route — recovers value and removes the hazard

Table 24. Sludge disposal and utilisation routes.

9.5 Chrome-bearing sludge: the special problem

Everything in that table turns on one question: does the sludge contain chromium? If it does, it is classified as **hazardous waste**, and the whole economics of disposal changes.

Two points must be clear.

- **Cr(III) and Cr(VI) are completely different in hazard.** Trivalent chromium, the form used in tanning, is fairly insoluble and of low toxicity. Hexavalent chromium is soluble, mobile, toxic and carcinogenic.
- **Careless handling can turn one into the other.** Cr(III) can be oxidised to Cr(VI) during incineration at the wrong temperature, in the presence of oxidising agents, or under alkaline conditions during long storage. A disposal route chosen without care can turn a manageable waste into a dangerous one.

The engineering conclusion is clear, and it returns to a theme running through the whole subject: chromium should be **segregated and recovered at the tannery**, in the primary stage, before it ever reaches the mixed sludge. Chrome recovered as basic chromium sulphate is a raw material worth money. Chromium spread through fifty tonnes of mixed sludge is a hazardous waste that costs money to bury.

KEY IDEA

The same chromium is either a recovered raw material or a hazardous waste. What decides which is a single upstream decision: whether the tanyard stream was segregated. Sludge disposal problems are created — or avoided — in the beamhouse and the tanyard, not at the landfill.

9.6 Reducing sludge at source

Sludge is expensive at every stage, so the cheapest sludge is the sludge never made. Four measures reduce it.

- **Chrome recovery** — removes the hazardous fraction and returns it to production.
- **Extended aeration** — endogenous respiration cuts biological sludge yield to a third or less of a conventional plant.
- **Anaerobic pre-treatment** — yields 0.05–0.15 kg VSS per kg COD, against 0.4–0.6 for aerobic treatment.

- **Cleaner production** — hair-save unhairing, high-exhaustion chrome tanning and lower chemical doses all cut the solids arriving at the plant.

9.7 Solid waste from the tannery itself

The tannery's other solid wastes are handled alongside the sludge. Fleshings, trimmings, splits and shavings are rich in collagen, and are increasingly seen as raw materials rather than wastes. Non-chrome trimmings and splits give gelatin and collagen hydrolysate; fleshings give fats and biogas; hair from hair-save unhairing gives keratin; chrome shavings can be treated with enzymes to separate protein from chromium. Each turns a disposal cost into a product, and each lightens the load on the landfill.

KEY POINTS — learn these

1. Treatment concentrates pollution into sludge; it does not destroy it.
2. The five stages are thickening, stabilisation, conditioning, dewatering, disposal.
3. Thickening is the most cost-effective step, because every later cost depends on volume.
4. The plate-and-frame press gives the driest cake and suits chrome sludge.
5. Cr(III) is low in toxicity; Cr(VI) is toxic and carcinogenic, and careless handling can create it.
6. Chromium must be segregated and recovered at the tannery, not extracted from mixed sludge.
7. Four ways to reduce sludge: chrome recovery, extended aeration, anaerobic pre-treatment, cleaner production.

REMEMBER THESE NUMBERS

Sludge handling is **a third to a half** of total plant operating cost.

Dry solids: primary **2–6%**, chrome **5–15%**, biological **0.5–2%**, chemical **1–3%**.

Thickening raises solids from about **1% to 4–6%**, cutting volume to about **one fifth**.

Cake solids: drying beds **25–40%**, belt press **18–25%**, plate-and-frame **30–45%**, centrifuge **20–30%**.

Lime stabilisation needs **pH above 12**. Anaerobic yield **0.05–0.15** kg VSS/kg COD against **0.4–0.6** aerobic.

LIKELY EXAM QUESTIONS

1. Why is it said that treatment concentrates rather than destroys pollution?
2. State the five stages of sludge processing in order, and say what each achieves.
3. Why is thickening the most cost-effective single step?
4. Distinguish Cr(III) from Cr(VI) in hazard, and state two circumstances in which one may become the other.
5. Why must chromium be segregated at the tannery rather than removed from the mixed sludge?
6. Name four measures that reduce the quantity of sludge produced.

Glossary of Symbols and Abbreviations

Term	Name	Meaning
AOP	Advanced Oxidation Process	A process that makes hydroxyl radicals to destroy hard organics
Breakpoint	Breakpoint chlorination	The dose beyond which free residual chlorine appears
CETP	Common Effluent Treatment Plant	A shared plant serving a cluster of industrial units
CT value	Concentration × time	The governing design quantity for disinfection
DBP	Disinfection By-Product	Compounds such as THMs formed by chlorination
GAC / PAC	Granular / Powdered Activated Carbon	The two forms of carbon used for colour and COD
Hazen	Hazen (Pt-Co) unit	Unit of true colour; ECR 2023 limit is 150
HOCl / OCl⁻	Hypochlorous acid / hypochlorite ion	The two forms of free available chlorine
Log removal	$\log_{10} (N_0/N)$	Logarithmic measure of disinfection performance
MF / UF / NF / RO	Micro-, ultra-, nano-filtration; reverse osmosis	The membrane family, in order of tightness
Permeate	—	The clean stream that passes through a membrane
Concentrate	also reject or brine	The rejected stream carrying the retained salts
Recovery	permeate flow ÷ feed flow	The fraction of the feed that comes out as permeate
Rejection	$(1 - \text{permeate} \div \text{feed}) \times 100$	The percentage of a solute held back by the membrane
THM	Trihalomethane	A chlorination by-product; suspected carcinogen
TDS	Total Dissolved Solids	Dissolved salts, mg/L; ECR limit 2,100
UVT	UV Transmittance	Percentage of 254 nm light passing through 1 cm of water
ZLD	Zero Liquid Discharge	A scheme in which no liquid effluent leaves the site
π	Osmotic pressure	In bar; about 0.077 bar per 1,000 mg/L of TDS
I × t	UV dose	In mJ/cm ² ; typically 20–40 for wastewater

Table 25. Terms used in this chapter.

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